

Engineering a Traceable Hybrid Controller for a Two-Wheeled Legged Robot

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Abstract—A two-wheeled legged robot couples pitch stabilization, commanded motion, leg posture and disturbance recovery. This project developed an inspectable control and evaluation system for HopperTrex in MjLab/MuJoCo: an identified classical controller with calibrated references, a fixed six-dimensional bounded residual interface, and version-bound qualification and evaluation records. The engineering contribution is the integration of these components, not a new reinforcement-learning algorithm. Historical experiments provide both positive and negative evidence. In a single-seed formal large-disturbance comparison, a residual policy reduced mean recovery time from 1.0130 to 0.8872 seconds, a 12.42% improvement over its matched classical baseline. Masking its leg channels reduced the relative improvement to 3.82% in a separately matched evaluation. A later affine gain-scheduled classical stack passed its flat-control qualification, but strict roll-over testing did not establish safe positive stair capability. Subsequent contact-backend and effort-control studies remain development diagnostics. This report closes the summer project with a reproducible evidence index, a preserved development snapshot and explicit limits: the results are simulation-only, the recovery claim is single-seed, and neither reliable stair climbing nor hardware transfer has been established.

Index Terms—wheel-legged robot, hybrid control, residual reinforcement learning, system integration, experiment provenance

I. MOTIVATION AND ENGINEERING OBJECTIVE

Two coaxial wheels must both balance an inverted body and produce motion. Leg posture changes the operating point at which that balance controller works. A successful implementation must therefore do more than maximize a training reward: it must preserve the meaning of commands, separate nominal control from learned corrections, and establish what each experiment actually measures. Classical balancing vehicles demonstrate the importance of state feedback and command decoupling [1]. Residual learning offers a way to retain a useful controller while learning a correction for unmodeled behavior [2].

The objective was to build and evaluate this division of responsibility on HopperTrex. The intended locomotion concept was wheel-driven motion with legs contributing posture regulation, not a permanently fixed-leg robot. Individual experiments nevertheless restricted authority to isolate questions. Formal RollBoundary used fixed posture references and zero policy residuals; its failure does not prove all active-posture methods impossible.

The engineering deliverables are a modular classical/residual control interface; a qualification chain binding results to

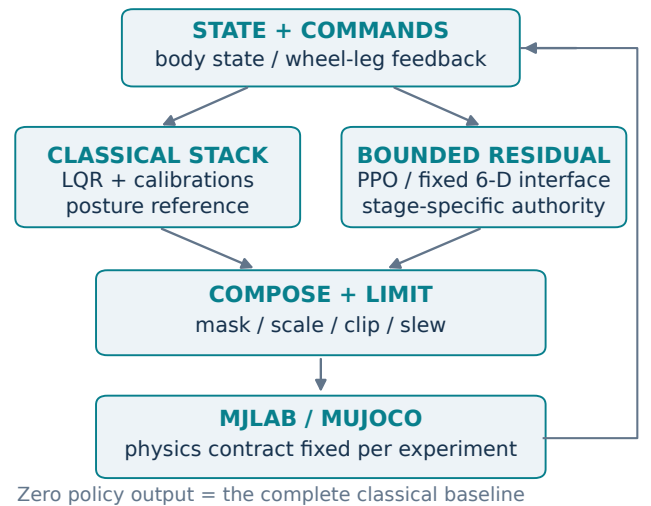


Fig. 1. Common system structure. The classical stack is experiment-specific: the later C1 affine gain schedule is not the earlier Stage5 recovery baseline.

controllers, checkpoints and protocols; and software interfaces with archived evidence for future work. Closeout recalculates existing measurements and checks selected software components. It does not extend the experimental campaign.

II. SYSTEM DESIGN

A. Simulation and Component Boundaries

The implementation uses MjLab on MuJoCo and its Warp-based simulation path. MuJoCo provides model-based dynamics [3]; parallel simulation is a practical way to collect reinforcement-learning experience [4]. Neither the simulator nor PPO is claimed as a new contribution. Project-specific work connects robot assets, task configuration, command generation, actions, training, evaluation and evidence packaging.

The robot has two wheel joints and four actuated leg joints. The conventional interface uses wheel velocity targets and leg position targets. The task layer owns observations, resets, rewards, curriculum state and metrics. A portable NumPy classical controller allows action conventions to be checked independently of a GPU rollout. The policy runner adds checkpoint validation and migration logic around the PPO implementation [5].

B. Classical Stack and Residual Composition

Let x_t denote the measured state, c_t the planar command and p_t^* the shaped height/pitch reference. The complete zero-residual controller is

$$u_t^{\text{cl}} = \pi_{\text{cl}}(x_t, c_t, p_t^*; \Theta, \Gamma), \quad (1)$$

where Θ identifies the balance controller and Γ its applicable calibration and posture artifacts: longitudinal calibration, yaw feedforward, posture-dependent station compensation, posture-to-joint mapping and reference rate limits. The matched baseline is the complete stack, not an isolated LQR term with other components removed.

The earlier Stage5 experiment used its frozen identified controller and calibrations. The later C1 campaign qualified an affine gain schedule,

$$u^{\text{bal}} = u_{\text{eq}}(h, p) - K(h, p)[x - x_{\text{eq}}(h, p)], \quad (2)$$

with parameters interpolated over a height/pitch grid. These are compatible architectural roles, not permission to relabel old results with a new controller. The applied target is

$$u_t = \mathcal{S}(u_t^{\text{cl}} + BM_k Da_t), \quad (3)$$

where a_t is a clipped policy output, D contains residual scales, M_k selects channels and B maps residuals to actuator targets. The operator $\mathcal{S}$ applies joint, wheel-speed and slew limits. These are engineering constraints, not a proof of closed-loop safety under every disturbance.

C. Fixed Six-Dimensional Contract

The actor channels retain the order

$$a_t = [r_{\text{bal}}, r_{\text{yaw}}, r_{\text{L,thigh}}, r_{\text{R,thigh}}, r_{\text{L,knee}}, r_{\text{R,knee}}]^\top. \quad (4)$$

After masking and scaling, the wheel corrections are

$$\Delta u_{\text{wheel}} = [-r_{\text{bal}} + r_{\text{yaw}}, r_{\text{bal}} + r_{\text{yaw}}]^\top. \quad (5)$$

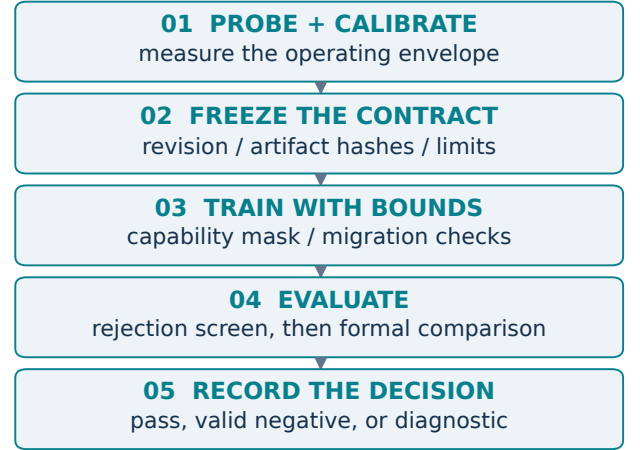
The remaining four channels add to nominal leg targets. Signs and order are part of the interface; they cannot silently change during migration. Capability masks vary by experiment, so dimensions alone do not establish active authority.

Stage5 recovery used wheel and leg residuals. Formal R0 used none. The later RollAssist route allocated residual authority to the four legs while retaining classical wheel control, but was not authorized by the failed R0 gate. The separate StairCamp campaign had its own contracts and was not RollAssist.

D. Reference Shaping and Migration

Posture references pass through a qualified map and rate limits before reaching leg servos. Static kinematic feasibility is insufficient: a posture can be geometrically possible but dynamically unsuitable. Zero-residual probes therefore establish usable envelopes before residual outcomes are interpreted.

Migration preserves channel semantics and binds policies to control artifacts. Actor exploration must also be inspected:



A completed run is not automatically a passed capability.

Fig. 2. Execution success, capability qualification and development diagnostics are separate states. A completed run does not automatically authorize promotion.

resuming a collapsed action distribution does not provide a useful start for every new capability. Curriculum state and provenance are checked rather than inferred from a filename. Routes requiring output-head zero initialization obtain an intended nominal-controller start, not a guarantee of subsequent learning progress.

E. Deployment Interfaces and Their Limits

The hardware abstraction layer separates joint/IMU observations, actuator commands, portable control and a latching safety supervisor. The synchronous deployment loop produces wheel velocity and leg position targets; its logs support later identification. No actual hardware session is reported.

The development snapshot adds optional effort telemetry, capability negotiation, a separate effort command and pure motor payload codecs. Missing capability, invalid telemetry or absent wheel-torque calibration must not silently fall back to another command mode. A separate effort prototype examines bilateral load allocation. It is not an already deployed replacement for the legacy loop. USB-CAN transport, physical motor calibration and integrated hardware validation remain unfinished.

III. VERIFICATION AND EVIDENCE ARCHITECTURE

A. Qualification Before Interpretation

Figure 2 summarizes the process. First measure the envelope and establish nominal control; then bind code, artifacts and protocol. Only an authorized route may train. A short screen can reject a candidate but cannot establish improvement. Formal comparisons use the registered metric and matched baseline; negative results remain in the record.

A formal record binds the task, revision, controller and calibration hashes, checkpoint identity where applicable, seed, profile and protocol. Hashes do not establish physical validity;

TABLE I
PRINCIPAL EVIDENCE USED IN THIS REPORT

ID	Revision	Role
E6/A1	2fb98ee	Recovery and leg ablation; formal, seed 1
C1	1f54968	Later affine-schedule flat qualification
C2-j3	f1d58e3	Detector qualification; valid negative
SC	ed1b9dd	StairCamp training-progress record
R0	0bc8df4	Strict roll boundary; formal, seed 1
R0c	d1d12e4	Dirty development snapshot with causal hashes; not formal

they make inputs and outputs identifiable. Raw trial and semantic checks remain necessary. For example, process success cannot establish gate success, and an upright pose does not establish the absence of unintended body contact.

B. Contact Timing and Failure Semantics

The roll-first protocol fixes a 0.02 s control period and four 0.005 s physics substeps. R0 monitors support from post-reset settling through success. Any bilateral unsupported substep permanently invalidates a trial; later traversal cannot retroactively restore success. A passing cell requires the registered success count and zero termination, non-wheel contact or bilateral unsupported events.

Sensor phase matters. Quantities associated with the constraint solve need not match those recomputed at the integrated endpoint. Later diagnostics explicitly distinguish these phases. Using one backend for dynamics and another only to override its support verdict changes the physical interpretation and is not an accepted shortcut.

C. Evidence Levels and Closeout Checks

The closeout separates implementation, formal evaluation, development diagnostics and unfinished work. Formal evidence can be negative and need not be a multi-seed aggregate. Table I identifies the principal revisions. Full hashes and paths are in the machine-readable manifest.

Closeout recalculated recovery ratios, checked the shared checkpoint identity, regrouped 480 R0 trials, inspected 15 C1 cells and checked 125 C2 outcomes. The effort sanity record's 16 causal source hashes match the archived code. A focused software check passed 45 tests and six subtests covering target composition, codecs, effort safety and pure control kernels. This does not mean the entire repository was retested or hardware safety was validated. No training or physics rollout was launched during closeout.

IV. REPRESENTATIVE RESULTS

A. Recovery Benefit and Leg Ablation

Stage5 evaluated one trained seed with 128 disturbance events per arm in its registered large-disturbance regime. Table II and Fig. 3 give stored mean recovery times. Improvement is

$$I = 100 \left(1 - \frac{T_{\text{candidate}}}{T_{\text{baseline}}} \right). \quad (6)$$

TABLE II
STAGE5 SINGLE-SEED FORMAL RECOVERY COMPARISON

Evaluation	Baseline (s)	Candidate (s)	Gain
Full residual	1.0130	0.8872	12.42%
Legs masked	0.9977	0.9595	3.82%

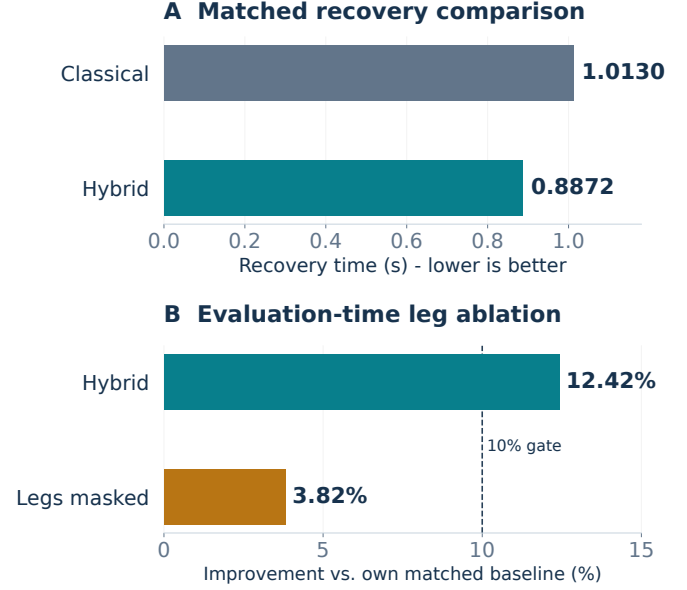


Fig. 3. Summary values exported from formal records, not invented trajectories. Each gain uses its corresponding baseline. Single-seed records do not support cross-seed error bars.

The full residual policy achieved 12.42%, exceeding the registered 10% requirement. Both arms recorded zero event terminations and zero non-wheel-contact event rates in this scenario.

Masking four leg channels at evaluation retains the same checkpoint but yields 3.82% relative improvement against its own matched baseline, below the 10% requirement. This supports a role for the legs in the tested benefit, not an exact percentage of causal contribution. The comparisons have slightly different baselines, and nonlinear interactions can change when channels are disabled.

The 128 events are not independent training seeds. No cross-seed confidence interval or general improvement claim is reported. A separate Stage1 retention record has an overall false verdict because its own improvement requirement was unmet; its non-improvement checks passed. The Stage5 recovery pass must not be rewritten as success on every gate.

B. Later Classical Qualification

C1 selected an affine gain-scheduled controller and passed 15 flat-control cells with no terminations or non-wheel contacts. The gate reports worst velocity error 0.00754 m/s, 95th-percentile absolute pitch error 0.01581 rad and 99th-percentile absolute pitch rate 0.18520 rad/s. These qualify the registered flat envelope, not stair climbing.

TABLE III
FORMAL R0 OUTCOMES ACROSS THE TWO POSTURE CARDS

Height	Successes	Interpretation
0 mm	96/96	Flat control qualified
2.5 mm	18/96	Both cards fail; 73/96 trials have unsupported samples
5 mm	0/96	Not qualified
7.5 mm	0/96	Not qualified
10 mm	0/96	Not qualified

The raw gate, selection file and original checksum list agree, whereas the old prose draft contained different summary numbers. This revision follows the hash-verified gate and records the correction in the evidence matrix. Original evidence and the old manuscript are preserved.

C. Strict RollBoundary Result

R0 tested two posture cards at five heights, with 16 environments and three repeats per card-height cell: 480 trials total, or 96 per height when cards are combined for display. Table III is recomputed from trial records. Combined counts are descriptive; qualification was separate for each 48-trial card cell, requiring at least 44 successes and zero safety events.

The largest tested common passing height is 0 mm and the first tested failing height is 2.5 mm. This brackets the tested transition but does not locate a continuous physical maximum. The record classifies no positive classical roll-over qualification and forbids training. Occasional successful traversals at 2.5 mm also mean it would be incorrect to say the robot never crossed. The controller did not establish the required safe capability.

V. CLOSING THE STAIR RESEARCH BRANCHES

The stair work changed control and physics contracts over time. It is a sequence of bounded investigations, not a single monotonic capability curve.

A. Detector and Residual-Camp Outcomes

C2-j3 completed 125 innovation-detector candidates over 288 paired trials and admitted none. Every candidate recorded zero timely detections under the registered requirement. This closes that route as a valid negative, not as proof that all contact detectors are impossible. The downstream C3/CEM chain did not produce a qualified classical climbing capability through that route.

A separately contracted StairCamp campaign subsequently ran. Its archived seed-1 progress record reports 1,000 completed updates, 20 curriculum evaluations and a final upper height still at 0.01 m. This is training-progress evidence, not completed multi-seed adjudication. Its final selection/evaluation chain was not completed in the records used here. The later RollAssist route must not be described as the same experiment.

B. Pose and Contact-Backend Diagnostics

After formal R0 failed, static posture screens and position-indexed schedules explored whether changing leg references

could provide a safe rolling path. They remained rejection-oriented diagnostics and did not replace the formal verdict. A synchronized-pose study was followed by native same-state contact replay and a separate native full-rollout study.

Explaining captured contact samples is different from qualifying a complete closed-loop system. Replay supported backend dependence at captured states, but full native trajectories could diverge and develop their own unsupported intervals. An alternative wheel collision representation also failed its promotion conditions. Backends or geometries therefore cannot be silently exchanged while claiming the same experiment.

C. Effort-Control Prototype

The latest development candidate made bilateral support load explicit and used a model-based effort kernel with leg impedance. Its sanity artifact records matching per-trial classifications in the two tested backends: the flat case passed and the 2.5 mm case safely stalled. The combined acceptance condition failed, so the larger registered reset set was not run. This small diagnostic does not establish that the general support problem is solved or isolate load allocation as beneficial.

Wheel-effort saturation and rapid command reversals motivate an inner-loop hypothesis, but the evidence contains no wheel-loop-only counterfactual that establishes a unique cause. Further controller design is future work, not a retrospective claim that the current prototype works.

VI. CONTRIBUTION, LIMITATIONS AND HANDOVER

A. What the Project Delivers

The principal contribution is a working separation of nominal control, bounded learning and evaluation responsibility. Task configuration, calibration, migration, action semantics and gate records are connected rather than treated as unrelated scripts. Portable control and testable deployment interfaces provide a basis for continuation. These are engineering deliverables, not a claim of algorithmic novelty over residual RL or PPO.

The student's role included research questions, experiment design, integration, diagnosis and evidence review. Development was collaborative and AI-assisted; sole authorship of every code path is not claimed. Identifying invalid measurements, retaining valid negative results and distinguishing simulator effects from control claims are part of the research outcome.

B. Limits of the Evidence

Recovery evidence is restricted to one training seed and one registered regime. Repeated events do not replace independent-seed replication. C1 and R0 use later controller versions and cannot be attached retrospectively to the recovery checkpoint. Contact outcomes depend on the physical contract and sampling phase. No hardware closed-loop performance, USB-CAN integration, robust friction-domain transfer or reliable stair-climbing capability is claimed.

The archive supports inspection of the displayed measurements and regeneration of figures without simulation. It does not promise bitwise reproduction on arbitrary future hardware. The development worktree contains uncommitted changes; it is

preserved as a snapshot, not a clean release. Passing selected software tests does not remove that distinction.

C. Handover and Future Work

The package contains revision/status records, a tracked diff and selected source snapshot, core evidence and checkpoint, a SHA256 manifest, extraction and plotting scripts, the poster and this report. Original evidence is unchanged. Documentation tools are isolated from the training environment. Closeout does not automatically merge, push or upload anything.

Continuation requires a newly registered question and budget: independent seed replication for recovery, a physically matched wheel-loop counterfactual, or guarded hardware bring-up with calibrated sensing and actuation. None is a prerequisite for the present FURP submission.

VII. CONCLUSION

HopperTrex has an integrated hybrid-control and evidence-management system with explicit authority, bounded residuals and version-bound evaluation. The archived single-seed recovery comparison measured 12.42% improvement, while the

leg-masked comparison fell below its improvement threshold. Later qualification and stair studies established limits rather than a general climbing capability. The project closes with demonstrated engineering functionality and traceable boundaries on what remains unproven.

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